Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	qwen3:8b

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Editorial Decision

MINOR REVISION

Weighted Score: 3.7 / 5.0
Confidence: 90%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	**5.0**	Excellent	■■■■■
Manuscript Structure	**2.0**	Weak	■■■■■
Methodology & Equations	**4.5**	Excellent	■■■■■
Statistical Analysis	**3.0**	Acceptable	■■■■■
Results & Evidence	**4.5**	Excellent	■■■■■
Discussion & Conclusions	**3.0**	Acceptable	■■■■■
Figures & Tables	**5.0**	Excellent	■■■■■
Scientific Writing	**3.5**	Good	■■■■■
References & Citations	**2.5**	Weak	■■■■■
Ethics & Transparency	**1.5**	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: **5.0 / 5.0** | Confidence: 100%

Strengths:

- [TITLE] Specific and informative, clearly reflects the contribution of the study.
- [ABSTRACT] Structured format with clear background, objective, methods, results, and conclusions.
- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.
- [ABSTRACT] Quantitative result signals are present in the abstract.

StructureReviewer

Manuscript structure and logical coherence

Score: **2.0 / 5.0** | Confidence: 90%

Strengths:

- Clear title and keywords section

Weaknesses:

- Missing abstract, references, title, and keywords sections
- Redundant content in section_vi and section_vii

Major Comments:

- The manuscript lacks critical sections such as abstract, references, title, and keywords, which are essential for a complete scientific paper.
- Redundant content in section_vi and section_vii suggests poor organization and logical flow.

Recommendations:

- Include the missing sections (abstract, references, title, and keywords).
- Revise sections_vi and section_vii to eliminate redundancy and ensure logical flow.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **4.5 / 5.0** | Confidence: 90%

Strengths:

- Clear problem definition and motivation
- Well-justified multi-objective optimization framework
- Statistical validation with nonparametric tests
- Reproducibility through detailed experimental setup

Weaknesses:

- No explicit methodology section, but inferred from the flow
- Limited detail on the exact neural network architecture
- No mention of hyperparameter tuning or sensitivity analysis

Major Comments:

- The method is well-justified but lacks explicit description of the neural network architecture and training procedure.

Minor Comments:

- More detail on the exact formulation of the multi-objective optimization problem would improve clarity.

Recommendations:

• Include a dedicated methodology section with detailed description of the neural network architecture and training procedure.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **3.0 / 5.0** | Confidence: 80%

Strengths:

• 30 independent runs are reported, which is a reasonable number for assessing robustness

• Bootstrap confidence intervals are reported, which is appropriate for non-parametric analysis

• Coefficient of variation, RMSE, MSE, MAE, and R^2 are reported and interpreted appropriately

• Pareto/hypervolume indicators are reported and interpreted as multi-objective performance metrics

Weaknesses:

• The Kruskal-Wallis test and post-hoc Wilcoxon/rank-sum tests are not reported with p-values or results, which is critical for assessing statistical significance

• The manuscript does not report the exact p-values for the Kruskal-Wallis and post-hoc tests, which is essential for interpreting results

Major Comments:

• The manuscript fails to report p-values for the Kruskal-Wallis and post-hoc Wilcoxon/rank-sum tests, which is a critical omission for assessing statistical significance.

Recommendations:

• Report the p-values for the Kruskal-Wallis and post-hoc Wilcoxon/rank-sum tests to fully assess statistical significance.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **5.0 / 5.0** | Confidence: 100%

Strengths:

• Equations are well-defined and numbered sequentially

• Tables and figures are referenced in text before appearance

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ResultsReviewer

Results validity and empirical support

Score: **4.5 / 5.0** | Confidence: 90%

Strengths:

• Clear presentation of MSE, RMSE, CV, and Pareto analysis

• Robust statistical validation with Kruskal-Wallis and Wilcoxon tests

Weaknesses:

• No explicit mention of statistical significance of the reported metrics

• No mention of computational time or efficiency in results section

Major Comments:

• The results section lacks explicit statistical significance reporting for the key metrics (MSE, RMSE, CV).

Recommendations:

• Include p-values or confidence intervals for the reported performance metrics to strengthen the validity of claims.

• Clarify the computational efficiency of the EANN framework relative to baselines in the results section.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.0 / 5.0** | Confidence: 90%

Strengths:

- Clear experimental validation
- Robust comparison with baselines

Weaknesses:

- No literature comparison in discussion
- Future work not specified

Major Comments:

- The Discussion lacks a comparison with prior work, which is critical for contextualizing the study's contribution.

Recommendations:

- Include a comparison with prior work in the Discussion section.
- Specify future work directions in the Conclusions section.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical description of methodology
- Well-structured abstract and introduction
- Robust statistical validation methods

Weaknesses:

- Overuse of passive voice
- Unclear acronyms
- Vague quantifiers

Major Comments:

- Passive voice overuse hinders clarity and readability
- Acronyms like EANN, PSO, and GA are undefined on first use
- Vague quantifiers like 'significant' and 'substantially' weaken the precision of claims

Recommendations:

- Define all acronyms on first use
- Revise passive voice constructions for clarity
- Quantify vague terms such as 'significant' and 'substantially'

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 90%

Strengths:

- Recent references from 2021-2023
- Diverse range of journals and conferences

Weaknesses:

- Lack of seminal foundational works
- Missing key areas in literature review

Major Comments:

• Missing seminal foundational works in structural dynamics and seismic design
• Key areas like structural health monitoring and seismic regulations are underrepresented

Recommendations:

• Include seminal works in structural dynamics and seismic design
• Add references to key literature on structural health monitoring and seismic regulations

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EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.5 / 5.0** | Confidence: 90%

Strengths:

• Limitations are explicitly stated.

Weaknesses:

• Limitations are not specific or detailed.

Major Comments:

• The manuscript lacks essential ethical reporting, including ethical approval, data availability, conflict of interest, and funding disclosure, which are critical for transparency and reproducibility.

Recommendations:

• Provide ethical approval details if human/animal studies were involved.
• Include a data availability statement.
• Declare any potential conflicts of interest.
• Disclose funding sources.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] The manuscript lacks critical sections such as abstract, references, title, and keywords, which are essential for a complete scientific paper.

[2] Redundant content in section_vi and section_vii suggests poor organization and logical flow.

[3] The method is well-justified but lacks explicit description of the neural network architecture and training procedure.

[4] The manuscript fails to report p-values for the Kruskal-Wallis and post-hoc Wilcoxon/rank-sum tests, which is a critical omission for assessing statistical significance.

[5] The results section lacks explicit statistical significance reporting for the key metrics (MSE, RMSE, CV).

[6] The Discussion lacks a comparison with prior work, which is critical for contextualizing the study's contribution.

[7] Passive voice overuse hinders clarity and readability

[8] Acronyms like EANN, PSO, and GA are undefined on first use

[9] Vague quantifiers like 'significant' and 'substantially' weaken the precision of claims

[10] Missing seminal foundational works in structural dynamics and seismic design

[11] Key areas like structural health monitoring and seismic regulations are underrepresented

[12] The manuscript lacks essential ethical reporting, including ethical approval, data availability, conflict of interest, and funding disclosure, which are critical for transparency and reproducibility.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] More detail on the exact formulation of the multi-objective optimization problem would improve clarity.

Specific Improvement Recommendations

1. Include the missing sections (abstract, references, title, and keywords).
2. Revise sections_vi and section_vii to eliminate redundancy and ensure logical flow.
3. Include a dedicated methodology section with detailed description of the neural network architecture and training procedure.
4. Report the p-values for the Kruskal-Wallis and post-hoc Wilcoxon/rank-sum tests to fully assess statistical significance.
5. Include p-values or confidence intervals for the reported performance metrics to strengthen the validity of claims.
6. Clarify the computational efficiency of the EANN framework relative to baselines in the results section.
7. Include a comparison with prior work in the Discussion section.
8. Specify future work directions in the Conclusions section.
9. Define all acronyms on first use
10. Revise passive voice constructions for clarity
11. Quantify vague terms such as 'significant' and 'substantially'
12. Include seminal works in structural dynamics and seismic design